

Sicheng Ouyang

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Skills

Languages: Python, TypeScript, Java, C/C++, SQL, JavaScript

Frameworks: FastAPI, React, Next.js, React Native, PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy

Infrastructure: PostgreSQL, MySQL, Redis, Supabase/PostGIS, Docker, Linux, GitHub Actions, Vercel, Tencent Cloud

Concepts: Data Structures & Algorithms, System Design, Database Design, Distributed Systems, Async Programming

Experience

Mui Scientific

SQL · REST APIs · ERP Integrations · Next.js

May 2026 – Aug 2026

Software Engineer

Canada · Mississauga

- Built SQL/REST integrations between enterprise resource planning and inventory systems, enabling serial/batch traceability and cutting audit turnaround by **87%**.
- Automated reconciliation across procurement, warehouse, and sales with validation and exception workflows, eliminating **80%** of manual inventory verification.
- Migrated the corporate site from PHP/WordPress to Next.js and standardized reusable UI components, improving accessibility, performance, and deployment maintainability.
- Rebuilt a Java Android control app for a clinical testing device, improving usability and maintainability.

KnowYi Technology Inc

Python · Next.js · OpenAI API · Async Pipelines

April 2025 – May 2025

Founding Engineer

Canada · Waterloo

- Built a Python/Next.js dashboard centralizing company records, AI pitch decks, website research, and real-time insights.
- Built source-linked OpenAI pipelines converting PDF, Excel, and CAD inputs into structured, reviewable spreadsheet data.
- Built async research pipelines delivering one-click website analysis in under **15 seconds** and replacing repetitive manual steps.

Tencent Music Entertainment Group

Python · Speech Synthesis · Audio ML

Jul 2024 – Aug 2024

Machine Learning Engineer

China · Shenzhen

- Ran **20+** experiments across **10** speech models to optimize generation and vocal-simulation configurations.
- Standardized cross-model evaluation with reusable Mel-spectrogram and F0 preprocessing workflows.
- Raised vocal consistency by **20%** and MOS from **3.43** to **3.63** through iterative model and inference tuning.

Advanced A-Level Study Website

WordPress · PHP · MySQL · Tencent Cloud

Sep 2023 – Aug 2025

Co-Founder & Principal Software Development Engineer

China · Wuhan

- Co-founded a learning platform serving **1,000+** daily active users and **200+** students over two academic years.
- Owned WordPress/PHP delivery, MySQL operations, and Tencent Cloud deployment for continuous publishing and maintenance.

Projects

The Cobalt Guide

Next.js · TypeScript · Supabase/PostGIS · MapLibre · Redis

[GitHub](#)

- Built a Canada-wide rewards platform with PostGIS geospatial search and MapLibre maps for nearby card-benefit discovery.
- Secured reporting with Supabase Auth/RLS and Redis rate limits, enforcing user-scoped access and abuse controls.
- Built a four-layer CI/CD quality gate covering linting, unit, E2E, and RLS tests, deployment, and comment-triggered performance checks.

Real-Time Chat Platform

FastAPI · Next.js/React · WebSocket · PostgreSQL

[GitHub](#)

- Designed a FastAPI/PostgreSQL backend with paginated REST and incremental WebSocket sync for persistent, real-time messaging.
- Implemented JWT auth, SMTP verification, rate limits, and structured API errors, blocking unverified access and improving operability.

Neural Network Playground

FastAPI · PyTorch · React · WebSocket

[Demo](#)

[GitHub](#)

- Built FastAPI training/evaluation APIs for a no-code ML playground spanning visual model design and **5** datasets.
- Implemented live loss visualization in React, turning backend updates into interactive model-training feedback.

Education

University of Waterloo, Canada

Sep 2025 – Present

Bachelor of Software Engineering